

Problem 1: Grocery Bill Calculator

Problem Statement

Ramesh visited a grocery store and purchased three items. The prices of the items are provided as input. Write a Python program to calculate the total bill amount.

The program should read the prices of the three items and display the total amount payable.

Input Format

The first line contains the price of Item 1.

The second line contains the price of Item 2.

The third line contains the price of Item 3.

Output Format

Display the total bill amount.

Constraints

$1 \leq \text{Item Price} \leq 10000$

Sample Input 1

500

180

60

Sample Output 1

740

Explanation

Total Bill = $500 + 180 + 60$
= 740

Python Template

```
def calculate_bill(price1, price2, price3):  
    # Write your code here  
    pass
```

```
if __name__ == "__main__":  
    price1 = int(input())  
    price2 = int(input())  
    price3 = int(input())  
  
    result = calculate_bill(price1, price2, price3)  
    print(result)
```

Problem 2: Student Percentage Calculator

Problem Statement

A student has appeared for five subjects. Calculate the total marks and percentage obtained by the student.

Assume that each subject is out of 100 marks.

Input Format

Five lines containing marks of five subjects.

Output Format

First line should display total marks.

Second line should display percentage.

Constraints

$0 \leq \text{Marks} \leq 100$

Sample Input 1

80
90
85
75
95

Sample Output 1

425
85.0

Explanation

Total = $80 + 90 + 85 + 75 + 95 = 425$

Percentage = $425/500 \times 100$
= 85.0

Python Template

```
def calculate_percentage(m1, m2, m3, m4, m5):  
    # Write your code here  
    pass  
  
if __name__ == "__main__":  
    m1 = int(input())  
    m2 = int(input())  
    m3 = int(input())  
    m4 = int(input())  
    m5 = int(input())  
  
    total, percentage = calculate_percentage(m1, m2, m3, m4, m5)  
  
    print(total)  
    print(percentage)
```

Problem 3: Employee Salary Calculator

Problem Statement

An employee receives a basic salary, house rent allowance (HRA), and bonus. Calculate the net salary.

Formula:

Net Salary = Basic Salary + HRA + Bonus

Input Format

First line contains Basic Salary.

Second line contains HRA.

Third line contains Bonus.

Output Format

Display the Net Salary.

Constraints

$1000 \leq \text{Salary} \leq 1000000$

Sample Input 1

30000

5000

2000

Sample Output 1

37000

Python Template

```
def calculate_salary(basic, hra, bonus):
```

```
    # Write your code here
```

```
    pass
```

```
if __name__ == "__main__":
```

```
    basic = int(input())
```

```
    hra = int(input())
```

```
    bonus = int(input())
```

```
    result = calculate_salary(basic, hra, bonus)
```

```
    print(result)
```

Problem 4: Electricity Bill Calculator

Problem Statement

An electricity board charges customers based on the number of units consumed.

Write a program to calculate the electricity bill amount.

Formula:

$$\text{Bill} = \text{Units} \times \text{Rate Per Unit}$$

Input Format

First line contains units consumed.

Second line contains rate per unit.

Output Format

Display the electricity bill amount.

Constraints

$$1 \leq \text{Units} \leq 10000$$

$$1 \leq \text{Rate} \leq 100$$

Sample Input 1

250

8

Sample Output 1

2000

Python Template

```
def calculate_bill(units, rate):
```

```
    # Write your code here
```

```
    pass
```

```
if __name__ == "__main__":
```

```
    units = int(input())
```

```
    rate = int(input())
```

```
    result = calculate_bill(units, rate)
```

```
    print(result)
```

Problem 5: Restaurant GST Bill

Problem Statement

A customer receives a restaurant bill. GST of 18% is applicable on the food amount. Calculate the GST amount and final bill amount.

Input Format

First line contains the food amount.

Output Format

First line displays GST amount.

Second line displays Final Bill.

Constraints

$1 \leq \text{Food Amount} \leq 100000$

Sample Input 1

1000

Sample Output 1

180.0

1180.0

Explanation

$\text{GST} = 1000 \times 18/100$
 $= 180$

$\text{Final Bill} = 1000 + 180$
 $= 1180$

Python Template

```
def calculate_restaurant_bill(amount):  
    # Write your code here  
    pass  
  
if __name__ == "__main__":  
    amount = float(input())  
  
    gst, final_bill = calculate_restaurant_bill(amount)  
  
    print(gst)  
    print(final_bill)
```

Problem 6: Travel Expense Calculator

Problem Statement

Rahul is planning a trip. He wants to calculate the total travel expense by adding transportation cost, hotel cost, and food cost.

Write a program to calculate the total expense.

Input Format

First line contains transportation cost.

Second line contains hotel cost.

Third line contains food cost.

Output Format

Display total travel expense.

Sample Input 1

500
2000
1000

Sample Output 1

3500

Python Template

```
def calculate_expense(travel, hotel, food):  
    # Write your code here  
    pass  
  
if __name__ == "__main__":  
    travel = int(input())  
    hotel = int(input())  
    food = int(input())  
  
    result = calculate_expense(travel, hotel, food)  
    print(result)
```

Problem 7: Mobile Recharge Cashback Calculator

Problem Statement

A telecom company offers cashback on mobile recharges. Given the recharge amount and cashback amount, calculate the final amount spent by the customer after cashback.

Input Format

The first line contains the recharge amount.

The second line contains the cashback amount.

Output Format

Display the final amount spent.

Constraints

$1 \leq \text{Recharge Amount} \leq 10000$

$0 \leq \text{Cashback} \leq \text{Recharge Amount}$

Sample Input 1

499

50

Sample Output 1

449

Explanation

Final Amount = Recharge Amount - Cashback

= 499 - 50

= 449

Python Template

```
def calculate_final_amount(recharge, cashback):
```

```
    # Write your code here
```

```
    pass
```

```
if __name__ == "__main__":
```

```
    recharge = int(input())
```

```
    cashback = int(input())
```

```
    result = calculate_final_amount(recharge, cashback)
```

```
    print(result)
```

Problem 8: Fixed Deposit Maturity Amount

Problem Statement

A customer invests money in a bank fixed deposit. Calculate the Simple Interest and Maturity Amount.

Formula:

Simple Interest = $(\text{Principal} \times \text{Rate} \times \text{Time}) / 100$

Maturity Amount = Principal + Simple Interest

Input Format

The first line contains the principal amount.

The second line contains the rate of interest.

The third line contains the time period in years.

Output Format

The first line displays the simple interest.

The second line displays the maturity amount.

Constraints

$1000 \leq \text{Principal} \leq 10000000$

$1 \leq \text{Rate} \leq 20$

$1 \leq \text{Time} \leq 30$

Sample Input 1

100000

7

2

Sample Output 1

14000.0

114000.0

Python Template

```
def calculate_fd(principal, rate, time):
```

```
    # Write your code here
```

```
    pass
```

```
if __name__ == "__main__":
```

```
    principal = float(input())
```

```
    rate = float(input())
```

```
    time = float(input())
```

```
    interest, amount = calculate_fd(principal, rate, time)
```

```
    print(interest)
```

```
    print(amount)
```

Problem 9: Online Shopping Bill Calculator

Problem Statement

An online shopping platform charges a delivery fee along with the product price. Calculate the final payable amount.

Input Format

The first line contains the product price.

The second line contains the delivery charge.

Output Format

Display the final payable amount.

Constraints

$1 \leq \text{Product Price} \leq 100000$

$0 \leq \text{Delivery Charge} \leq 1000$

Sample Input 1

1500

100

Sample Output 1

1600

Explanation

Final Amount = Product Price + Delivery Charge
= 1500 + 100
= 1600

Python Template

```
def calculate_bill(price, delivery):  
    # Write your code here  
    pass
```

```
if __name__ == "__main__":  
    price = int(input())  
    delivery = int(input())  
  
    result = calculate_bill(price, delivery)  
    print(result)
```

Problem 10: Rectangular Room Area Calculator

Problem Statement

A civil engineer wants to calculate the area of a rectangular room.

Formula:

$$\text{Area} = \text{Length} \times \text{Width}$$

Input Format

The first line contains the length of the room.

The second line contains the width of the room.

Output Format

Display the area of the room.

Constraints

$$1 \leq \text{Length} \leq 1000$$

$$1 \leq \text{Width} \leq 1000$$

Sample Input 1

20

15

Sample Output 1

300

Explanation

$$\text{Area} = 20 \times 15$$

$$= 300$$

Python Template

```
def calculate_area(length, width):  
    # Write your code here  
    pass
```

```
if __name__ == "__main__":
```

```
    length = int(input())
```

```
    width = int(input())
```

```
    result = calculate_area(length, width)
```

```
    print(result)
```

Problem 11: Fuel Cost Calculator

Problem Statement

A vehicle owner wants to calculate the total fuel cost for a trip.

Formula:

Fuel Cost = Litres Consumed × Cost Per Litre

Input Format

The first line contains the litres of fuel consumed.

The second line contains the cost per litre.

Output Format

Display the total fuel cost.

Constraints

$1 \leq \text{Litres} \leq 500$

$1 \leq \text{Cost Per Litre} \leq 200$

Sample Input 1

35

105

Sample Output 1

3675

Explanation

Fuel Cost = 35×105
= 3675

Python Template

```
def calculate_fuel_cost(litres, price):  
    # Write your code here  
    pass
```

```
if __name__ == "__main__":  
    litres = int(input())  
    price = int(input())  
  
    result = calculate_fuel_cost(litres, price)  
    print(result)
```